

1 Flexible Shaft: Student Guide



This document provides a fundamental project overview with generic procedures for conducting the Flexible Shaft experiment utilizing LabVIEW to measure and control a dynamic system. The LabVIEW program presented herein was developed by instructional staff as a ready-to-use utility.

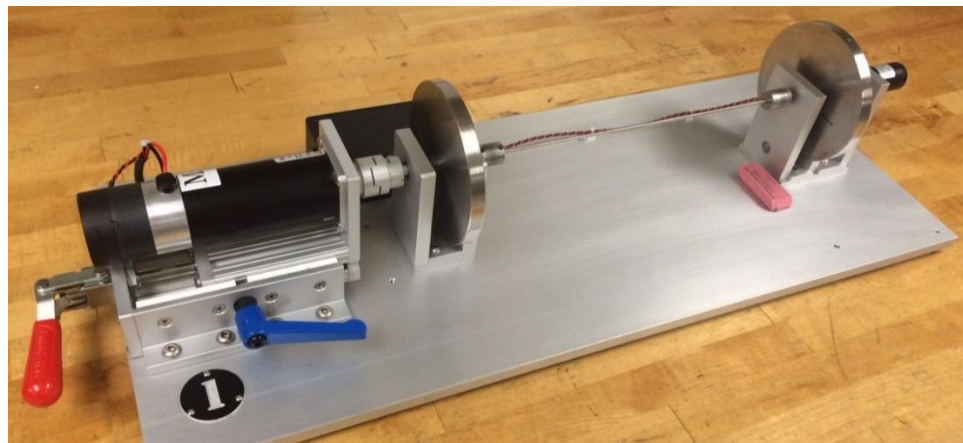


Figure 1-1: Close-up of Flex Shaft test fixture

1.1 Project Overview

The ME495 project examines the dynamics of two flywheels separated by a long, thin shaft. An electric motor drives the proximal flywheel, with the response of the distal flywheel measured by a tachometer. A MATLAB/Simulink model is developed and validated using the test fixture. All system variables, such as damping coefficients and the motor constant, are experimentally determined using various methods. The LabVIEW program provides two methods for determining the system's frequency response: Frequency Sweep or Random Noise. Hardware wiring depends on the selected method (see the wiring schematic). Data from both methods may be saved as a tab-delimited text file, which may be easily imported into Microsoft Excel or MATLAB for analysis.

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1.2 Equipment

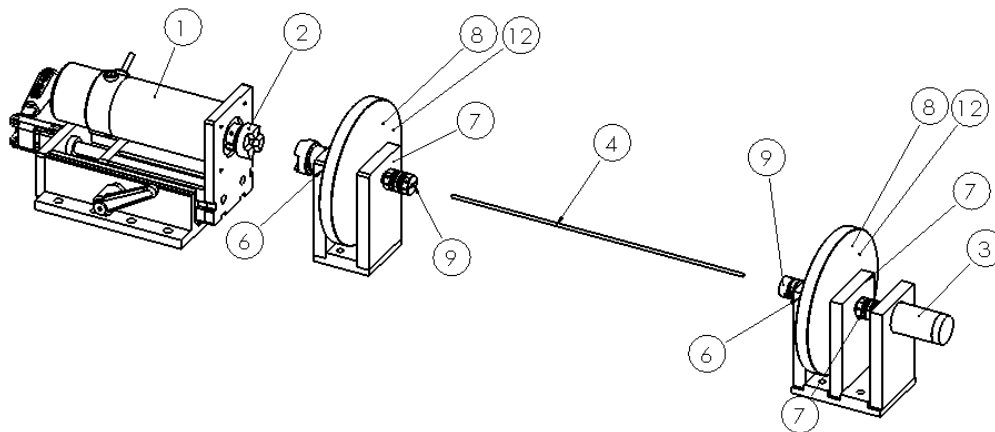
- ☐ Flexible Shaft test fixture with:
 - *MagMotor* DC motor with tachometer (P/N: C21-E-300-F-T7)
 - *Servo-Tek* Tachogenerator (P/N: SA-740B-1)
 - Two flywheels coupled by a long, small-diameter shaft
 - Interface box with 0.1 ohm (10W, 1%) current shunt resistor
- ☐ Keysight DSOX-1102A digital oscilloscope
- ☐ Keysight 33509B function generator
- ☐ Techron 7520 power supply amplifier
- ☐ National Instruments (NI) cDAQ Chassis with USB interface
 - NI 9229 – Analog Input Module
 - Analog BNC Breakout box
- ☐ BNC cables

- (4x) 24"
- (3x) 36"
- (4x) BNC 'Tee' splitter
- (2x) BNC-to-Banana jack adapter
- USB 2.0 cable – Type-A to B
- Pink Eraser [McMaster: 13225T21] for locking the flywheel in place
- LabVIEW running *Flex Shaft Main.vi*



Figure 1-2: General setup for the experiment

1.2.1 Key Components/Replacement Parts:



Item#	Description	Part No.	Vendor
1	DC motor with tachometer	C21-E-300-F-T7	MagMotor
2	Tapered Teeth Coupling; 6061 Aluminum	Custom	UM
3	Servo-Tek Tachometer (distal)	SA-740B-1	Servo2Go Ltd.
4	1/8" x 12" Rod; Air-Hardened Tool Steel; D2	88565K1	McMaster-Carr
5	Magnetic Switch	65985K15	McMaster-Carr
6	Flanged Bearing; ABEC-7; ID: 3/16"; OD:1/2"	4262T14	McMaster-Carr
7	Double Shield Bearing; ABEC-7; ID: 3/16"; OD:1/2"	3759T35	McMaster-Carr
8	Fenner Transtorque-Mini Coupling (6410019)	80817943	MSC Direct
9	Helical Flex Coupling; 3/16" to 1/8"	6208K451	McMaster-Carr
10	Helical Flex Coupling; 3/8" to 3/16"	6208K493	McMaster-Carr
11	Shaft; 3/16" Diam. X 4"; 316 Stainless Steel	1263K154	McMaster-Carr
12	Flywheel; 304 Stainless Steel	Machined	UM

Table 1-1: Experimental Fixture Component List.

1.3 Engage (Couple) Motor

Step 1: Verify power to the motor is OFF.

Step 2: Rotate the blue 'L-handle' CCW to unlock the linear slide.

Step 3: Actuate the toggle clamp (red handle) to advance the motor slide.

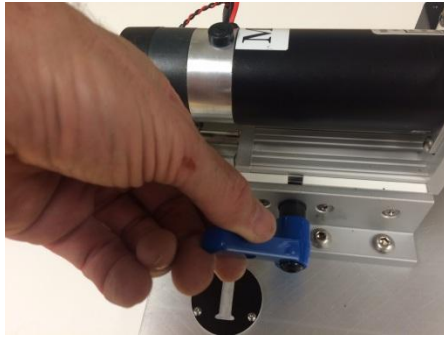


Figure 1-3: Unlock the linear slide



Figure 1-4: Actuate toggle clamp

Step 4: Verify the coupling teeth are oriented to mate.

Step 5: Fully advance the motor slide into the coupling

Step 6: Rotate the blue 'L-handle' CW to lock the linear slide.



Figure 1-5: Teeth not properly oriented.

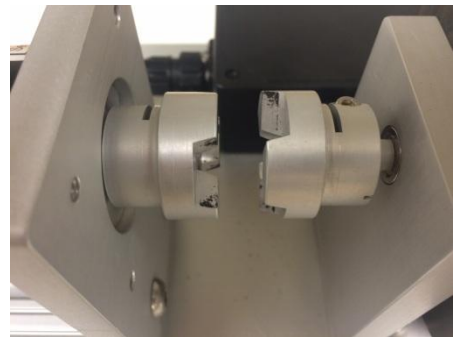


Figure 1-6: Teeth oriented to mate.

Note: DO NOT FORCE ENGAGEMENT! The motor slide should easily advance. If not, inspect the fixture and/or consult with the GSI.

1.4 Setup for Manual Data Acquisition

1.4.1 Fixture:

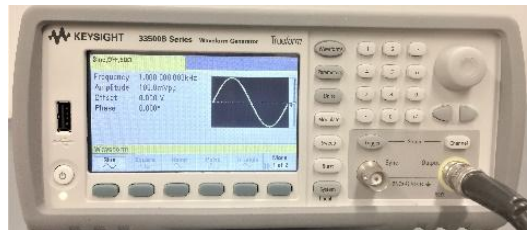
- Step 1: Connect the red & black banana cables to the **+VDC** & **COM** socket, respectively, located on the interface box of the flexible shaft fixture. The other end of the banana cords connects to the **+** & **-** Channel 1 output terminals located in the rear of the Techtron 7520 power supply amplifier.
- Step 2: Use a 24-inch BNC cable to connect the **Voltage** terminal on the interface box to **Channel 1** of the oscilloscope.
- Step 3: Use a 24-inch BNC cable to connect the **Distal Tach** terminal on the interface box to **Channel 2** of the oscilloscope.



Figure 1-7: Flexible shaft interface box

1.4.2 Function Generator:

Step 1: Connect the **Output** of the function generator to the **Channel 1** input terminal at the rear of the Techron 7520 power supply amplifier.



Step 2: Sinusoidal waveform:

- Press the **WAVEFORMS** button (right of the screen).
- Press the **SINE** softkey (assigned on screen).



Step 3: Set waveform parameters:

- Press the **PARAMETERS** button (right of the screen).
- Select the **FREQUENCY** softkey and set it in the range of 4 to 10 Hz using the numeric keypad or encoder knob.
- Select the **AMPLITUDE** softkey and set to $200\text{mV}_{\text{p-p}}$.
- Select the **OFFSET** softkey and set to zero.

Note: This setting may vary among setups; the objective is to produce a ~ 4 Vp-p sine wave. Display the power-supply output on the oscilloscope to verify the signal. Begin with the power amplifier set to zero, then slowly increase its output until the signal reaches 4 Vp-p or the amplifier reaches full power, whichever occurs first. If necessary, adjust the function generator's amplitude

until the oscilloscope displays a 4 Vp-p signal with the power amplifier at full power.

Step 4: Set output termination to High-Z:

- Press the **CHANNEL** button (above the **Output** BNC connector).
- Toggle the **OUTPUT** softkey to **ON**.
- Press the **OUTPUT LOAD** softkey, then select **SET TO HIGH-Z** softkey.
- Press the **DONE** softkey to exit.

Step 5: Connect the USB cable between the function generator (rear port) and the PC.

1.4.3 Oscilloscope:

The following oscilloscope settings provide a baseline signal measurement. These settings are merely suggestions and may be modified/tweaked to best capture the desired signal.



Step 1: **Channel 1** (Motor Voltage) set to 10.00 V/div:

- Use **Volts/Div** knob.

Step 2: **Channel 2** (Distal Tach) set to 500 mV/div:

Step 3: **Time** axis set to 50 ms/div.

Step 4: Set to **Roll** mode:

- Press the **Acquire** button (located below the Horizontal label).
- Press the **Time Mode** softkey and select **Roll**.

Step 5: Trigger Type: Edge.

Step 6: Trigger Source: 1.

Step 7: Slope: Rising Edge.

Step 8: Trigger Mode: Auto.

Step 9: Verify the Probe ratio is 1:1:

- Press the Channel button (**1** or **2**).
- Navigate softkeys **Probe (Next)** >> **Probe**.
- Verify attenuation is set to **Ratio**.
- Use the **Entry** knob to set the ratio to **1:1**.

Step 10: Press the **Run/Stop** button to initiate acquisition (should be green).

Note: If banana-to-BNC connectors are utilized in the setup, the ground side must be maintained throughout the circuit. Reversing the signal polarity may cause erroneous readings (magnitude will be incorrect).

1.5 Setup for Automated Data Acquisition

The oscilloscope is optional in this setup. Instead, the motor voltage and tachometer-generator signals are displayed along with the motor current in LabVIEW. If the oscilloscope is to remain in the experimental setup, two BNC splitter connectors will be required to provide the voltage and tach signals to the NI-9229 Analog Input module on the cDAQ 9172/9178 chassis.

Step 1: Connect a BNC cable from the **Voltage** terminal of the interface box to **AI0** of the NI-9229 Analog Input module.

Step 2: Connect a BNC cable from the **Current** terminal of the interface box to **AI1** of the NI-9229 Analog Input module.

Step 3: Connect a BNC cable from the **Distal Tach** terminal of the interface box to **AI2** of the NI-9229 Analog Input module.

Step 4: Connect a BNC cable from the Motor Tach terminal of the interface box to AI3 of the NI-9229 Analog Input module [OPTIONAL].

Step 5: [OPTIONAL] If the oscilloscope is desired, use a BNC splitter at the cDAQ interface and connect a BNC from AI0 (Motor Voltage) to Channel 1, AI2 (Distal Tach) to Channel 2 of the oscilloscope.

Step 6: Connect a BNC cable (or banana patch cords) from the Ch1 Output of the Techron power amplifier to the motor voltage supply on the interface panel (+VDC & COM) using a BNC-to-Banana plug adapter at both ends.

1.5.1 Frequency Sweep Method

- Step 1: Connect a USB (Type-A to B) cable from the PC to the function generator (rear port).
- Step 2: Connect a BNC cable from the **Output** port (front) of the function generator to **Ch 1. Input** port of Techron power amplifier (rear).

1.5.2 Random Noise Method

This method uses a 'random' excitation waveform generated in the LabVIEW program to drive the power amplifier. The function generator is not utilized and should be disconnected from the system during this experiment.

- Step 1: Connect a BNC cable from **A00** of the NI-9263 Analog Output module to **Ch 1. Input** port of Techron power amplifier (rear port).

1.6 LabVIEW Codes

The automated procedure requires the LabVIEW program *Flex Shaft Main.vi* – Version 2011.01 or newer. If the version or date is not displayed, please check with the GSI to verify that the most recent program is utilized.

The **ME x95 LabVIEW Shortcuts** folder is installed on all users' desktops. This folder contains shortcuts to all LabVIEW programs used in ME 395/495. Double-click the folder icon to view all the lab shortcuts. Select the *Flex Shaft Main* shortcut to open up the LabVIEW program.

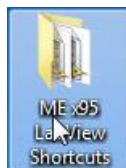


Figure 1-8: Shortcuts folder desktop icon

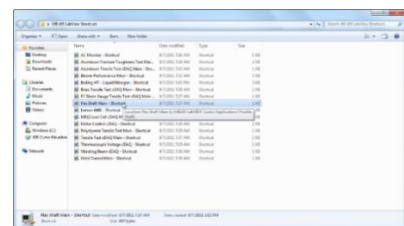


Figure 1-9: Shortcuts folder

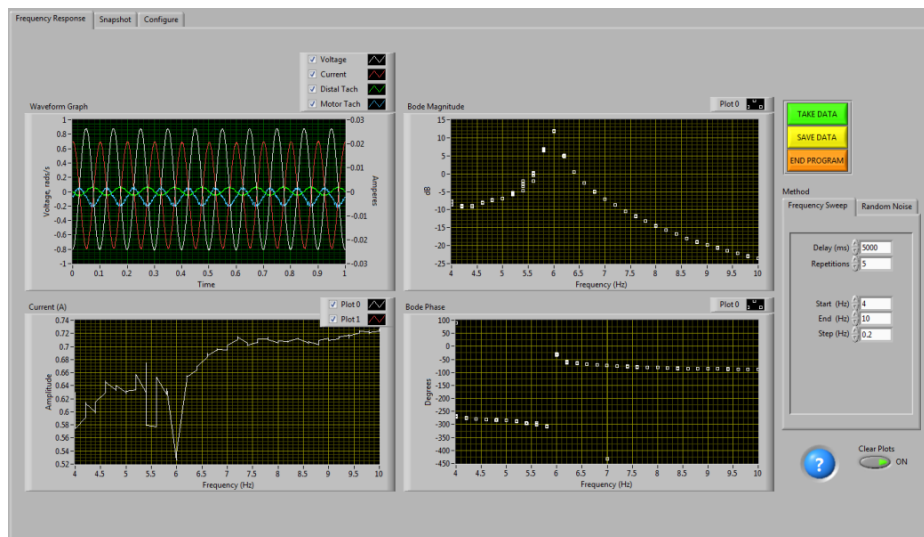


Figure 1-10: Front Panel of Elite LabVIEW program.

1.6.1 Configuration:

Two tabs are located in the upper-left region of the front panel: **Take Data** and **Configure**. Click on the **Configure** tab to view or edit the configuration settings. The default settings/channels should be correct. For convenience, the default settings are listed below:

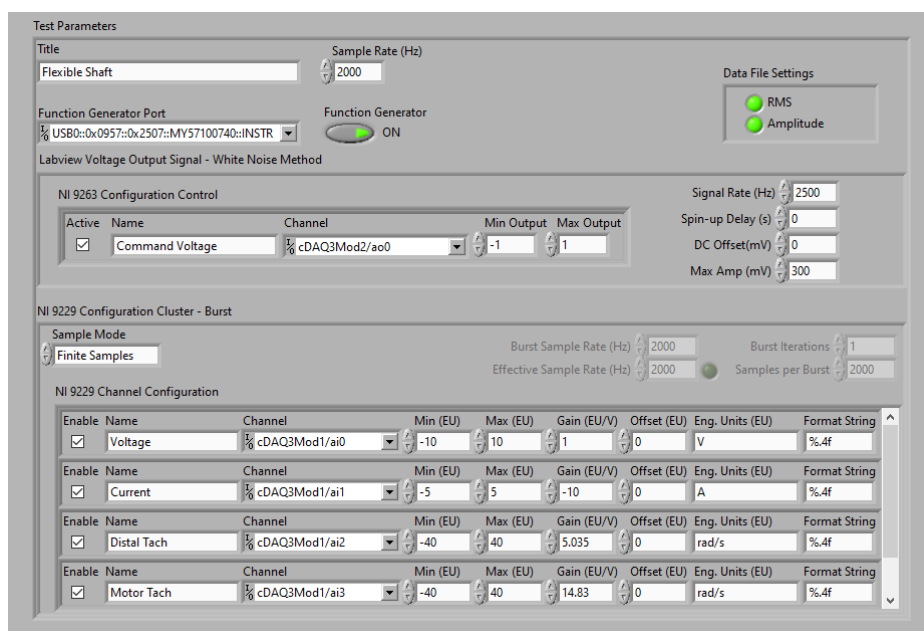
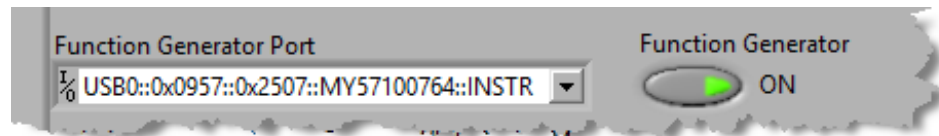


Figure 1-11: Default configuration and settings.

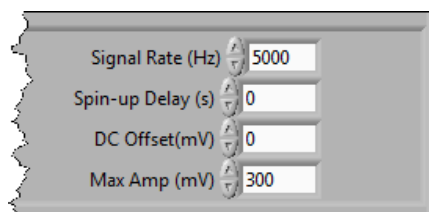
1.6.1.1 Function Generator

The function generator should be enabled (i.e., toggled ON) when using the Frequency Sweep method to determine the system's frequency response. Additionally, the power amplifier should be connected to the **Output** of the function generator (on the front panel). An error may occur if the function generator is enabled during a random noise test.



1.6.1.2 Random Noise Voltage Output Parameters

The following parameters allow the user to modify the output voltage signal to the power amplifier during a random noise test. If a **DC Voltage** is desired, a sufficient **Spin-up Delay** is recommended to allow the motor to reach steady-state speed before introducing the noise signal and recording the data. The **Signal Rate** determines the resolution of the output signal to the power amplifier. The **Max Amp** scales the output signal to limit the peak voltage to prevent damage to the shaft.



1.6.2 Snapshot

This feature was added for FA2012 to enable the collection of raw data for analysis. Typically, an oscilloscope is used to measure several system variables. This panel provides another tool that collects the raw data in digital format at a specified sampling rate and duration, which is extremely useful for determining the damped natural frequency and other system parameters.

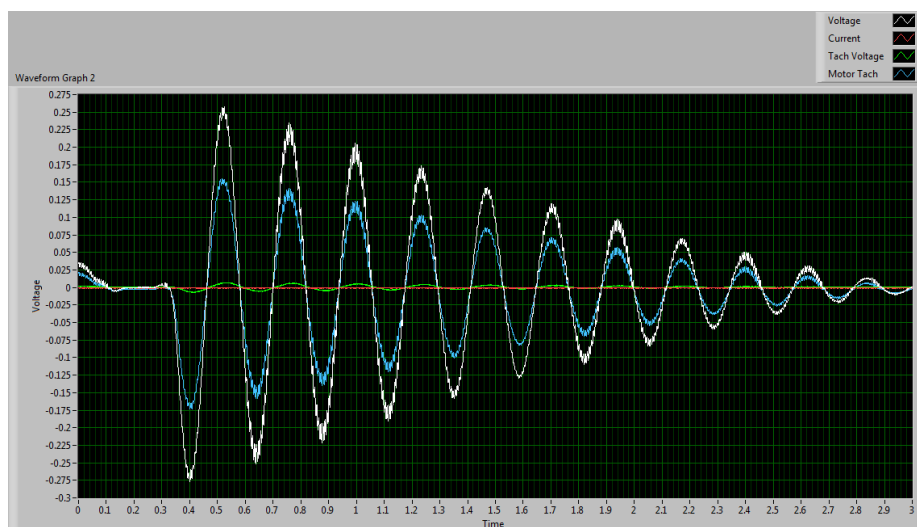
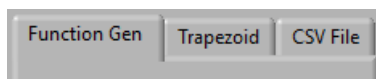


Figure 1-12: Snapshot panel for collecting raw data.

1.6.3 Motor Excitation

Motor excitation may optionally be configured in the **Snapshot** mode with three methods available.



The power amplifier wiring will need to be modified depending on the selected excitation mode. In the **Function Gen** mode, the Keysight function generator is controlled remotely. However, in the **Trapezoidal** and **CSV File** modes, LabVIEW outputs an analog voltage signal. In all cases, a small voltage signal is boosted by the Techron power amplifier to drive the motor. Please use extreme caution and verify the wired connections before executing a test.

Note: The total amplitude of the signal (Amplitude and DC Offset) must not exceed 300 mV!

Warning: Excessive motor excitation may permanently damage the shaft. In extreme cases, the slender shaft can shatter. Use discretion when utilizing the motor excitation utility.

1.6.3.1 Function Generator

Select the **Function Gen** tab to remotely control the function generator to excite the motor with a standard, cyclic waveform. Verify the **Output** of the Keysight function generator is wired to the **Ch-1 Input** of the Techron power amplifier (see wiring diagram).



Figure 1-13: Output terminal on the function generator.

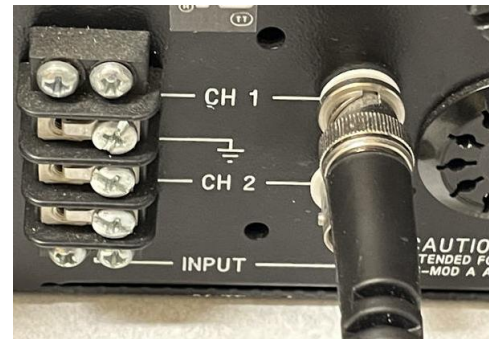


Figure 1-14: CH-1 Input on Techron amplifier.

In LabVIEW, the function generator must be enabled by toggling the rocker switch on (**Configuration** panel). The **Enabled** indicator displays the current status of the function generator. On the **Snapshot** tab, use the configuration frame to select the desired waveform (Sine, Square, Ramp, or Triangle), and set the associated parameters (i.e., frequency, amplitude, and DC offset). For tests that do not use the function generator, such as when determining the damped natural frequency, the function generator must be toggled off on the Configuration panel; otherwise, an error will occur.

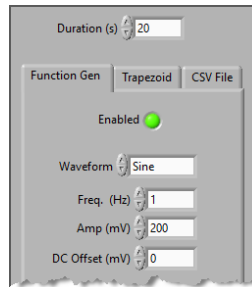


Figure 1-15: Function Gen configuration.

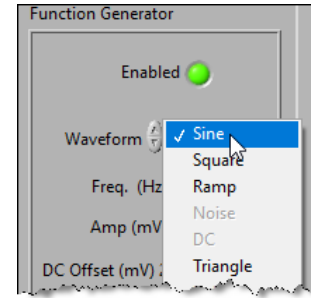


Figure 1-16: Available waveforms.

Note: At high sampling rates, the memory may max out before the desired test duration is attained. For longer-duration tests (20-30 seconds), use a slower sampling rate.

1.6.3.2 Trapezoidal Waveform

Since a trapezoidal waveform is not available on the Keysight 33509B function generator, the voltage signal is generated within LabVIEW. As such, the **AO0** analog output (cDAQ chassis) must be wired to the Techron power amplifier. Typically, the BNC cable is disconnected from the **Output** of the Keysight function generator and connected to the **AO0** analog output on the cDAQ chassis. LabVIEW will output a voltage signal to the power amplifier, which drives the motor.



Figure 1-17: Analog output (AO0).

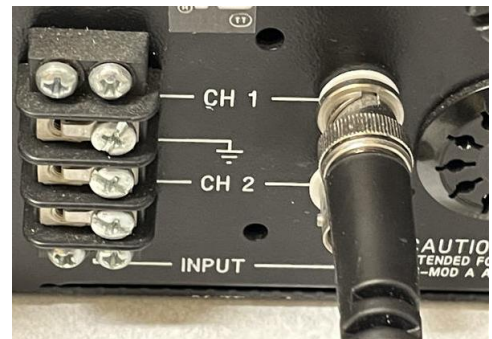


Figure 1-18: Techron power amplifier - CH 1 Input.

In LabVIEW, the function generator must be disabled by toggling the rocker switch off (**Configuration** panel). On the **Snapshot** screen, select the **Trapezoid** tab and use the associated frame to configure the desired trapezoidal waveform (frequency (Hz), amplitude (mV), and DC offset). The ramp rate of the trapezoid is defined by the **Ramp Factor**, which can vary from zero (square wave) to one (pure triangle wave). Due to singularity issues, zero is not permitted. For a pure square wave, use the function generator.

To preview/edit the trapezoidal waveform, toggle the **Show Waveform** button. The waveform will be displayed in the adjacent **Waveform Graph**

2. Toggle the button again to hide the waveform and return to normal mode.

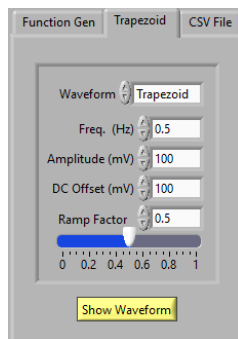


Figure 1-19:
Trapezoidal.

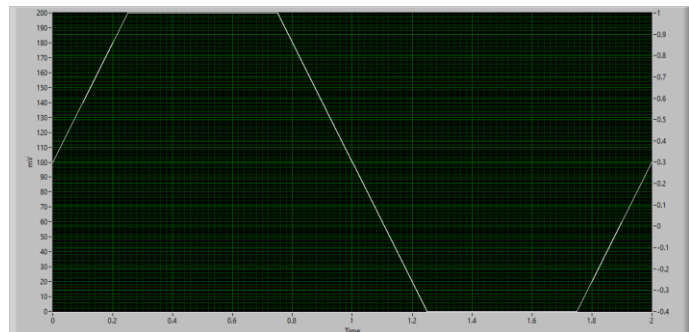


Figure 1-20: Preview waveform.

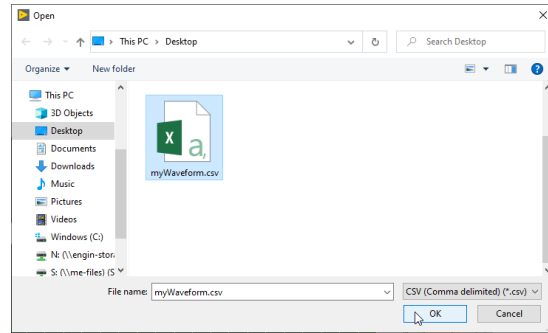
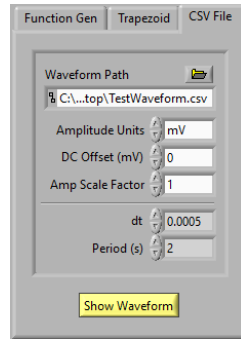
Note: While in preview mode, the **Take Data** button is disabled. Click the **Hide** button to return to normal operating mode.

1.6.3.3 Arbitrary Waveform (CSV file)

The **CSV File** tab enables a comma-delimited text file containing the arbitrary waveform definition to be imported into the LabVIEW utility. Similar to the **Trapezoidal** waveform, an analog voltage signal is generated within LabVIEW to drive the motor. The **AO0** analog output (cDAQ chassis) must be connected to the Techron power amplifier [See Figure 1-17 and Figure 1-18].

The comma-delimited data file can be generated using any software (Microsoft Excel, MATLAB, Python, Octave, etc.). The text file must contain only two columns: time and amplitude – no column headers. The time series must consist of a sequence of amplitude values at successive equally spaced points in time. The amplitude may be in units of volts or millivolts. The time series defines a single cycle. The defined cycle will be repeated over the test duration. If the test duration is less than the defined cycle, the test cycle will be clipped. The imported cycle is automatically resampled within LabVIEW to generate a signal with the sampling rate defined on the **Configuration** panel. So, if the imported waveform file only has 100 data points over a three-second cycle, LabVIEW will generate a signal with 7500 points (assuming a sampling rate of 2500 Hz) using linear interpolation.

Select the file icon to open a browser and navigate to the desired CSV data file. Use the **Amplitude Units** dropdown to select the appropriate units of the defined waveform. The calculated time increment (**dt**) and duration of the arbitrary cycle (i.e., **Period**) are displayed. An optional **DC Offset (mV)** can be applied without recreating the waveform. Additionally, the amplitude may be scaled using the **Amp. Scale Factor**.



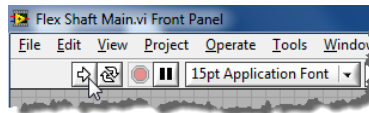
The **Show Waveform** button may be toggled to view the arbitrary waveform as currently defined. While in preview mode, the **Take Data** button is disabled. Click the **Hide** button to return to normal mode.

Note: The total amplitude of the signal (Amplitude and DC Offset) must not exceed 300mV! The signal is automatically clipped to prevent damage to the equipment.

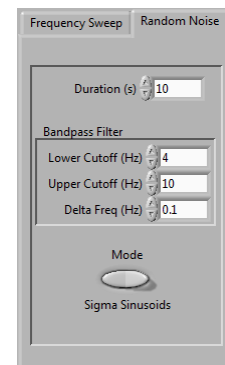
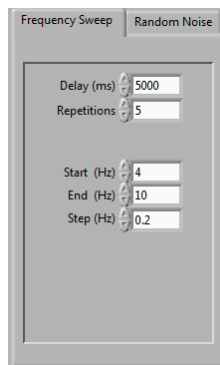
1.6.4 Frequency Response

Click on the **Frequency Response** tab to view the main panel. The LabVIEW program will automatically determine the frequency response of the system by constructing the Bode diagrams based on the distal tachometer (output) and motor voltage (input). The user may select between the frequency sweep or random noise method using the appropriate tabbed sub-panel.

Step 1: Start LabVIEW: Click on the white arrow located along the top menu bar.



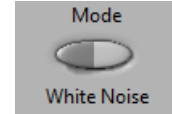
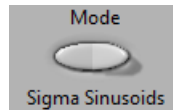
Step 2: Select the desired method and set parameter values.



Frequency Sweep sequentially steps through a user-defined frequency range. At each frequency, the test may be replicated by a defined number of repetitions, thereby providing multiple samples with a pause (or delay) between frequency increments to allow the system to attain a steady state.

Random Noise generates a multitoned signal to excite the motor over a user-defined duration. The excitation signal may be generated by two modes: band-limited white noise or the summation of sinusoids (Sigma Sinusoids). The latter combines a

series of sinusoidal waves, each shifted by a randomized phase angle. Both methods are effective for producing Bode diagrams, each with its own set of pros and cons.



Step 3: Press the **Take Data** button to initiate the test.

The DC motor will automatically be cycled through the defined test.



Step 4: When the test concludes, a file window will be displayed to save the data to a tab-delimited file for analysis.

- **Stop** or **Save Data** will terminate program execution and save the accumulated data to a text file.

Step 5: Press **End Program** to exit the program.



The program should always be ended by pressing the **Stop** button. Do not abort the program using the stop sign in the toolbar or by clicking the red 'X' in the upper right corner of the program window. Doing so will stop the program, but the motor may continue operating in its last controlled state.

1.7 Procedures

1.7.1 Estimating Friction Coefficients

- Step 1: Turn the Techron power amplifier OFF.
- Step 2: Disconnect power to the motor (**+VDC & COM**).
- Step 3: Lock flywheel #1 using the pink eraser.
- Step 4: Give flywheel #2 a slight twist.
- Step 5: Record the free response on the oscilloscope (roll mode is recommended) – or use the **Snapshot** utility.
- Step 6: Compute the ratio **r**. This calculation should be performed using the experimental response over several cycles to reduce the effects of noise and other uncertainties. Once **r** is known, calculate **ζ**.
- Step 7: Calculate the friction coefficient **Btr**.
- Step 8: Design a test procedure to obtain the friction coefficient **Btl**.
(Move the BNC cable to **Motor Tach**, lock flywheel #2, and twist flywheel #1).
- Step 9: [OPTIONAL] Design a test procedure to determine all three coefficients **Bshaft**, **Bmotor**, and **Bbearing**. If you choose to skip this step, you can assume **Bmotor** = 0 and attribute half of **Btr** to **Bshaft** and the other half to **Bbearing**. If you choose an alternative method for separation, please justify it.

1.7.2 Determine the Frequency Response of the System

Use an oscilloscope to measure the magnitudes of the input voltage **Vin(t)** and the output angular velocity. Also, measure the relative phases between the two signals. Before you do, keep this in mind: The shear stress in the shaft is related to **Vin**.

Warning: Do NOT exceed V_{p-p} excitation voltage or you will weaken and possibly snap the shaft

Use the following procedure in taking your frequency-response measurements:

- Step 1: Set the amplifier gain to its lowest setting (full CCW).
- Step 2: Turn **ON** the function generator and press the **WAVEFORMS** button, then the **SINE** softkey (below screen, first key on the left).
- Step 3: Press the **AMPLITUDE** softkey on the function generator. Set the value to ~ 200 mVp-p.
- Step 4: Press the **FREQUENCY** softkey on the function generator. Set the frequency of the function generator to 1 Hz.
- Step 5: Turn ON the equipment and observe the motor voltage on the oscilloscope.
- Step 6: Increase the Techron amplifier gain control very slowly until the peak-to-peak motor voltage is between 3-4 volts. Don't exceed an input voltage of 4 Vp-p. On the other hand, use reasonably large inputs to reduce the effect of noise/uncertainties.
- Step 7: Measure the response for input frequencies between 1 and 20 Hz. Use intervals of 1 Hz. Near resonance, take measurements at least every 0.2 Hz (because the resonance region is quite narrow due to very low damping, make sure you don't skip it in your measurements). Measure the current across the frequency range, but pay special attention to it at higher frequencies.

Note: May simply press the **PARAMETERS** button to load the softkeys: **FREQUENCY**, **AMPLITUDE**, **OFFSET**, and **PHASE**.

1.7.3 How to Generate a DC Signal (Keysight 33509B)

- Step 1: Turn down (CCW) the amplifier control knob to zero.
- Step 2: Press the **WAVEFORMS** button on the function generator, which assigns various waveforms and options to the six softkeys.
- Step 3: Select the **MORE** softkey (rightmost) to advance menu options.
- Step 4: On the second menu, select the **DCV** softkey.
- Step 5: Set the offset to +0.100 VDC as a starting point (if the display is in millivolts, set it to ~100 mVDC). You may need to adjust this value to attain the desired speed.
- Step 6: Turn up (CW) the Techron amplifier control knob (gain) to initiate shaft rotation. The shaft should spin continuously with the oscilloscope displaying a flat trace at about 2V on Channel 1.
- Step 7: You can adjust the DC voltage level using the amplifier.

Use the Techron amplifier to provide fine adjustment (i.e., tweak) the applied motor voltage, which provides a much finer resolution than the function generator.

1.7.3.1 To Return to a Sinusoidal Input:

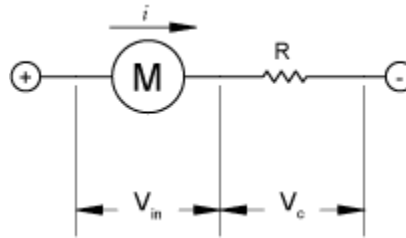
- Step 1: Turn the amplifier control knob to zero (full CCW).
- Step 2: Press the **WAVEFORMS** button.
- Step 3: Select the **SINE** softkey, which returns the unit to sinusoidal mode while retaining the previously set offset.
- Step 4: Set the **OFFSET** to zero.
- Step 5: Turn the amplifier back up to the determined maximum (4Vpp)

1.7.4 Measuring Current on Oscilloscope

A 0.1Ω (1%) current shunt resistor is in the black interface box. Motor current is calculated by measuring the voltage across the shunt resistor (V_c) and applying Kirchhoff's law:

$$I = \frac{V_c}{R} \quad (0.1)$$

The **Current** BNC interface provides a voltage signal across the shunt resistor, which may be measured on an oscilloscope. To provide a common reference for both motor voltage and current, the shunt is wired backward. Therefore, the oscilloscope signal should be inverted.



1.7.5 Sign Convention

All test fixtures were wired to produce voltage and tachometer signals in compliance with the *Right-Hand Rule* sign convention, as shown in Figure 1-21. The thumb is aligned parallel to the shaft, pointing outward from the motor to the distal flywheel. The fingers curl in the positive direction of rotation. Therefore, a positive voltage input to the motor results in positive rotation, as monitored by both tachometers.

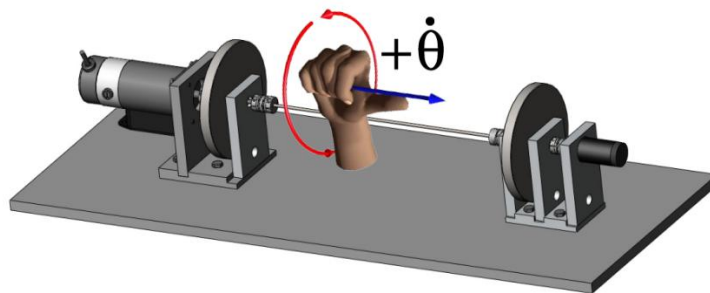


Figure 1-21: Right-hand rule sign convention applied to the fixture.

1.8 Sample Data

1.8.1 Screenshots

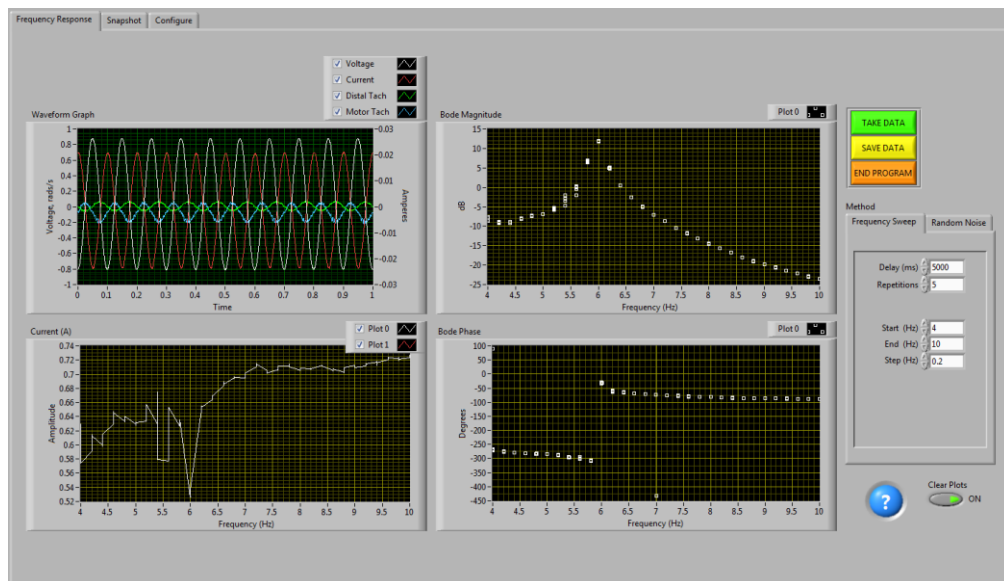


Figure 1-22: Sample screenshot – Frequency Sweep method.

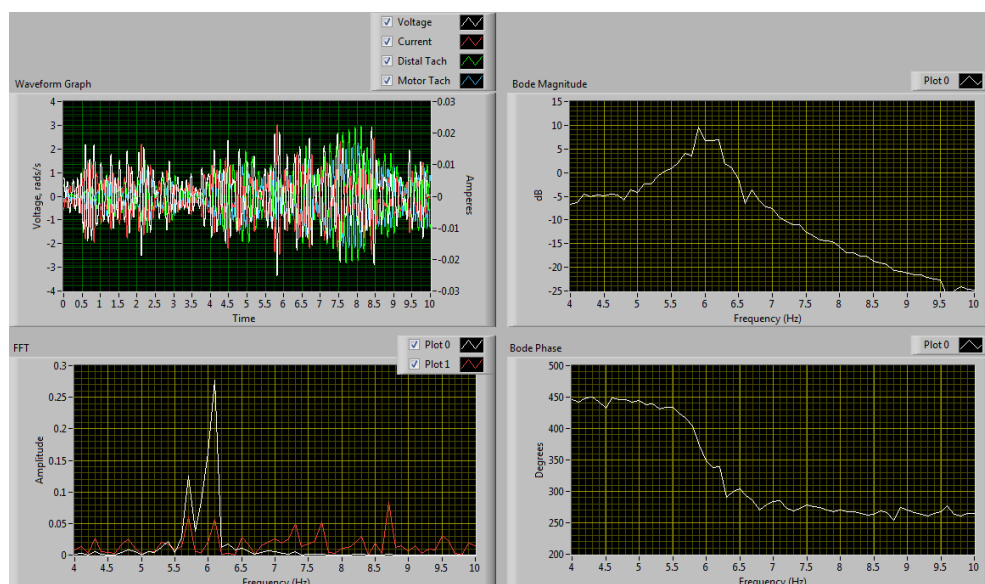


Figure 1-23: Frequency Response utilizing Random Noise method (band-limited white noise).

1.8.2 Data Files

The following are sample listings from the data files. These tab-delimited files may be imported into Microsoft Excel or MATLAB for analysis.

1.8.2.1 Frequency Sweep Method

	A	B	C	D	E	F	G	H
1	Flexible Shaft							
2	1/15/2024 8:17							
3	Sample Rate (Hz) = 2000.00							
4	Effective Sample Rate (Hz) = 2000.00							
5	Frequency (Hz)	Current (A)	Bode Magnitude (dB)	Bode Phase (deg)	Voltage (V_rms)	Current (A_rms)	Distal Tach RMS (rad/s)	Motor Tach RMS (rad/s)
6								
7	3.9998	1.0974	-6.8332	-90.3375	1.8123	0.7901	0.8314	0.1074
8	4.5001	1.2787	-5.1069	-96.8486	1.7951	0.8998	0.9985	0.1718
9	5	1.2618	-2.5377	-104.0697	1.7945	0.8937	1.3478	0.6013
10	5.5	1.2265	3.5051	-114.3563	1.7924	0.8704	2.6917	2.083
11	6	1.0441	7.542	-226.4458	1.8068	0.7412	4.3187	4.8094

Figure 1-24: Sample frequency sweep data file opened in Microsoft Excel

Note: If enabled on the Configuration tab, the amplitude values of the analog channels are displayed in successive columns similar to the RMS values.

1.8.2.2 Random Noise Method or Snapshot

	A	B	C	D	E
1	Flexible Shaft				
2	7/31/2020 9:48				
3	Sample Rate (Hz) = 2000.00				
4	Effective Sample Rate (Hz) = 2000.00				
5	time (s)	Voltage (V)	Current (A)	Distal Tach (rad/s)	Motor Tach (rad/s)
6					
7	0	1.5302	-0.2883	7.0649	7.0692
8	0.0005	1.5312	-0.2793	6.9583	7.0721
9	0.001	1.532	-0.2749	6.9041	7.0699
10	0.0015	1.5331	-0.2673	6.9931	7.0586
11	0.002	1.5341	-0.2599	7.0582	7.0367
12	0.0025	1.5357	-0.2531	6.9887	7.0162
13	0.003	1.5369	-0.2378	6.9142	7.0119

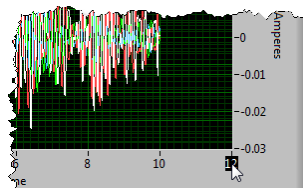
Figure 1-25: Sample Snapshot-mode data file opened in Microsoft Excel

1.9 LabVIEW Tips

1.9.1 Change Axis Limits

Step 1: Double-click on the upper (or lower) limit

Step 2: Edit the value



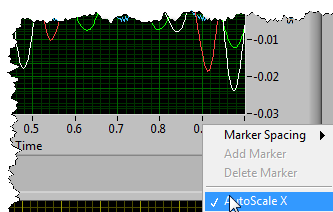
1.9.2 Toggle AutoScale

AutoScale is the default setting for all plot axes, which automatically sets the axis limits to best display the entire dataset. The user may wish to zoom up on a range and not have to repeatedly reset the axis limits. To enable (or disable) AutoScale:

Step 1: Move the cursor to the desired plot or axis.

Step 2: Right-click the mouse button.

Step 3: Select **AutoScale X** (or **AutoScale Y**).



1.9.3 Toggle Plot Visibility

Use the checkboxes in the plot legend to toggle the visibility of the desired channel.

